# Lochinvar EE Induction Heater Controller

Designed By: Aaron Neuharth, Austin DuCrest, John Donnell, Cole Wilson, and Dow Cox

## Abstract

This project presents the design and prototyping an induction coil controller in such a way to heat both the internal and surface level material of a circular heating element. The system utilizes an inverter bridge to generate magnetic fields, inducing eddy currents within a circular metal workpiece, causing contact-less heating. By integrating a closed-loop control system, the controller provides both PWM generation as well as precise data collection. This design establishes a framework for an efficient induction heating system, with future development focusing on scaling to a higher voltage as well as PID tuning for a more accurate heating process.

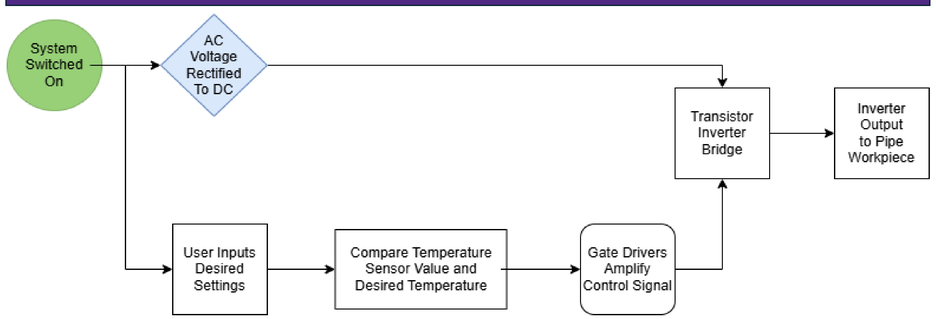
## Problem Statement

Design and prototype an induction coil controller in such a way to heat both the internal and surface level material of a circular heating element.

## Project Overview Objectives

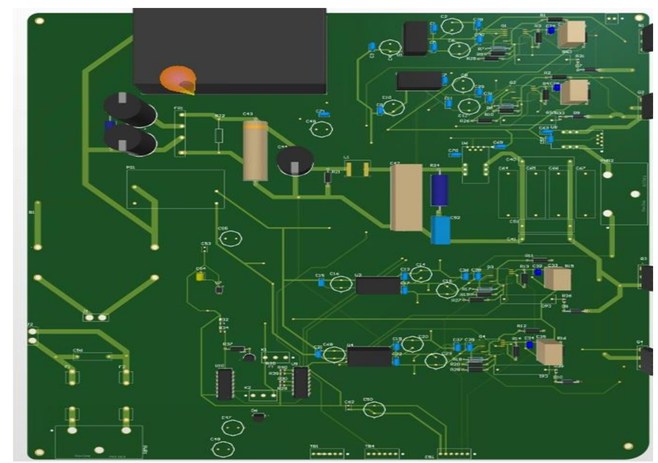
- Designed to have multiple power settings
- Induce surface eddy currents and internal heating
- Measure total electrical power
- Measure total temperature rise of a metal pipe.

## Logic Flow Chart




## Bill of Materials

| | |
|---|---|
| Power Subsystem | - **$200** |
| Safety Subsystem | - **$500** |
| PCB Subsystem | - **$500** |
| Controls/Embedded System | - **$130** |
| Total Project Budget | - **$1330** |



## Chosen Subsystems & Technical Details

Our design was separated into four different subsystems: **Power Subsystem, Safety Subsystem, PCB Subsystem, and the Embedded/Controls Subsystem**. For the **Power Subsystem**, a full bridge inverter was selected to maximize power delivery to the induction coil while maintain a high efficiency. The **Safety Subsystem** involves a hardware-level latching fault circuit designed to trigger a shutdown in the event a fault occurs. The **PCB Subsystem** includes a multi-layered board that was designed to handle not only the full-load voltage, but also the heat generated by the IGBTS. Finally, the **Embedded/Controls Subsystem** utilizes an STM32 microcontroller to produce our PWM signals as well as handle our PID control. This system connects the rest of the systems together and is what enables the system to induce surface eddy currents through the coil.

## Experimental Analysis

- Gate driver Circuit produced 30kHZ square waveform when given a PWM input.
- Safety Latch Circuit tested to latch when fault is detected.
- Controller successfully able to read input from thermocouples with an accuracy around ±0.5C.

## Future Goals

Future work for this project will focus full-load testing of the system to measure the thermal rise of the workpiece. This includes going from a 30VDC power supply to a 120VAC supply. Additionally, future experiments should compare full-load power input versus thermal measurement to improve PID control.

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